

Natural human anti-Gal alpha(1,3)Gal antibodies react with human mucin peptides.

We have recently demonstrated that both antibodies to Gal alpha(1,3)Gal, and the Gal alpha(1,3)Gal binding lectin (IB4), bind a synthetic peptide (DAHWEWL), there being a similar recognition of carbohydrate and peptide structures. We now report that the anti-Gal alpha(1,3)Gal antibodies and IB4 lectin also react with peptides encoded by mucin genes (MUC 1, 3, 4)-sequences known to be rich in serine, threonine and proline. This activity was demonstrated (1) by the ability of mucin derived peptides to block the reaction of anti-Gal alpha(1,3)Gal antibodies and IB4 lectin with a Gal alpha(1,3)Gal+ pig endothelial cell line; the reactions were specific and did not occur with a random peptide containing the same sequences or with other mucin peptides; (2) by the fact that anti-mucin1 antibodies could react with the Gal alpha(1,3)Gal expressed after transfection of COS cells (Gal alpha(1,3)Gal-, Muc1-) with cDNA encoding the pig alpha, 3galactosyltransferase; and (3) that the IB4 lectin and anti-Gal alpha(1,3)Gal antibodies could react with mucin 1 found on the surface of human breast cancer cells. Thus natural occurring anti-Gal alpha(1,3)Gal antibodies found in all human serum can react with self (Muc1) peptides expressed in large amounts on the surface of tumour cells but not on normal cells. The findings are of interest and serve to explain the previously reported findings that human cells can, at times, express Gal alpha(1,3)Gal; such expression is an artefact, the reaction is due to the phenomenon described herein, i.e. that anti-Gal alpha(1,3)Gal antibodies react with mucin peptides.